

DISHA GOYAL

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SUMMARY

Data Engineer with hands-on experience building Databricks data pipelines using PySpark, Spark SQL, Delta Lake, JDBC ingestion, medallion architecture, Unity Catalog, and Databricks Workflows. Strong background in data cleaning, validation, dashboard-ready modeling, and AI/BI-ready datasets through academic research, internships, and end-to-end portfolio projects.

EDUCATION

University of Southern California: **Master of Science in Computer Science (GPA 3.75/4.0)** Aug 2024 - May 2026
Vellore Institute of Technology: **Bachelor of Technology in Computer Science (GPA 9.29/10)** Sep 2020 - May 2024

TECHNICAL SKILLS

- **Languages & Processing:** Python, SQL, PySpark, Spark SQL, Apache Spark
- **Data Engineering:** ETL/ELT pipelines, JDBC ingestion, batch processing, incremental ingestion, CDC concepts, data cleaning, data validation, metadata-driven pipelines, medallion architecture
- **Databricks & Lakehouse:** Databricks Workflows/Jobs, Lakeflow Connect, Lakeflow Jobs, Lakeflow Spark Declarative Pipelines, Auto Loader, Delta Lake, Delta tables, Unity Catalog, Secret Scopes, Databricks SQL
- **Pipeline Development:** COPY INTO, CTAS, MERGE INTO, streaming tables, rescued data column, data quality expectations, job triggers, task dependencies, failure handling, monitoring
- **Databases, BI & Developer Tools:** MySQL, MariaDB, dimensional modeling concepts, Excel, Power BI, Tableau, Git, GitHub, Jupyter Notebook, VS Code, CI/CD fundamentals, DevOps fundamentals

PROFESSIONAL EXPERIENCE

Institute for Creative Technologies, USC (Graduate Research Assistant - Data Engineering) Jan 2026 - May 2026

- Cleaned and validated WhisperX-generated transcription and speaker-diarization outputs for 100+ multi-issue bargaining negotiation sessions, preparing structured datasets for downstream conversational AI and negotiation-behavior research.
- Converted raw AI-generated conversation outputs into standardized speaker-wise transcript records across session ID, speaker label, timestamp, and utterance fields.
- Corrected speaker-attribution errors, missing dialogue turns, inaccurate text, and formatting inconsistencies to improve dataset quality, reliability, and usability for AI/ML research workflows.

Cosasco, Santa Fe Springs, CA (Engineering Intern - Data Infrastructure) Sept 2025 - Jan 2026

- Extracted and analyzed engineering test data from Anotrol 2020 CPS using SQL and MariaDB on Raspberry Pi, supporting reliability assessment, troubleshooting, and predictive maintenance workflows.
- Developed repeatable database export procedures and technical instructions for RDU test data, standardizing data extraction steps and reducing manual dependency for future analysis.
- Maintained structured test-set tracking, procedure mapping, and engineering documentation to support test infrastructure, data traceability, and new product enablement workflows.

Blackcoffer Pvt. Ltd., Delhi, India (Backend and Data Analytics Intern) Jan 2024 - Jul 2024

- Supported backend data operations for client-facing analytics dashboards by preparing structured datasets, validating data flows, and assisting with SQL-based API/data integration tasks.
- Organized relational client data for reporting and internal analytics, troubleshooting data availability and validation issues with backend, dashboard, and frontend teams to improve analytics reliability.

PROJECTS

Finance Domain End-to-End Data Engineering Pipeline (Databricks, PySpark, SQL, Delta Lake)

- Built an end-to-end Databricks lakehouse pipeline for finance data using PySpark, Spark SQL, JDBC ingestion, Autoloader-based incremental ingestion, and Delta Lake, following Bronze, Silver, and Gold medallion architecture.
- Developed reusable transformation and validation logic to clean, model, and load source data into analytics-ready Delta tables, including curated Gold-layer datasets for BI, reporting, and AI/ML consumption.
- Orchestrated workflows using Databricks Jobs, nested dependencies, Secret Scopes, Unity Catalog, and email alerts, enabling governed access, monitoring, and business-ready consumption through Dashboards and Genie.

U.S. EPA Emissions Analytics Dashboard (Databricks SQL, Data Visualization)

- Built Databricks SQL workflows to ingest, clean, and transform EPA emissions data for state and county-level analysis, including numeric type conversion and population-based calculations.
- Created dashboard-ready datasets and visualizations to identify high-emission regions, compare normalized emissions trends, and support geographic emissions analysis.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- **Databricks Academy Badges** - Associate Data Engineering and Data Analyst Learning Paths
- Databricks Certified Data Engineer Associate - In Progress